

Component Purchase List - Exact Specifications

IoT-Based Automated Tropical Terrarium - AIUB, Faculty of Engineering, EEE4103, Summer 2025-26

Hand this sheet to the shop. Quantities include spares. Lines in orange are the variant traps - the wrong version either does not work or destroys the controller. SHOP column: R = RoboticsBD (Uttara) T = TechShopBD B = BDTronics D = Daraz online K = Katabon market / any components shop.

A. CORE BUILD - buy all of these

QTY	ITEM	EXACT SPECIFICATION	SHOP
1	ESP32 DevKit V1 development board	ESP-WROOM-32 module, 38-pin, dual-core 240 MHz, 4 MB flash, onboard PCB antenna, micro-USB, CP2102 or CH340 USB chip. MUST be WROOM-32 - NOT WROVER, NOT ESP32-S3/C3.	T / R
1	GY-BME280 sensor module	3.3 V version ONLY (marked GY-BME280 3.3V). Temperature + humidity + pressure, I2C interface, address 0x76 or 0x77, 4-pin (VCC GND SDA SCL). Do NOT accept the 5 V board.	R
1	GY-302 BH1750FVI light sensor module	Digital lux sensor, I2C address 0x23, 16-bit, 1-65535 lx range, 3.3 V, 5-pin (VCC GND SCL SDA ADDR).	R
3	Capacitive soil moisture sensor v1.2	Capacitive type (long PCB paddle, no exposed metal prongs), analog output, operates at 3.3 V, 3-pin (VCC GND AOUT). Two used + one spare.	R
1	Horizontal float switch, 52 mm	PP plastic, side-mount horizontal type, reed contact, 2-wire, rated min 0.5 A / 100 V DC. Mounting thread approx. M10.	R
2	Ultrasonic mist maker module, 5 V	20 mm / 113 kHz atomizer disc + driver PCB kit, 5 V DC input approx. 0.5 A. MUST start misting the moment power is applied - NO touch button. Reject any module that needs a button press to start.	B / D
1m	LED strip, SMD 5050, 12 V, cool white	12 V DC, SMD 5050 chips, 60 LED per metre, cool white 6000-6500 K, 1 metre length, IP20 or IP65.	D
1	F5305S 4-channel MOSFET switch board	4-channel, trigger input 3-20 V (3.3 V logic OK), output 5-36 V DC, 5 A per channel continuous. Common DC input, 4 switched outputs.	R
2	D4184 MOSFET switch module	AOD4184 N-channel module, trigger 2.5-20 V, load 6-36 V DC up to 10 A, PWM capable. One used for the grow light + one spare.	R
1	DC power adapter, 12 V 5 A	Input 100-240 V AC, output 12 V DC 5 A regulated switching adapter, 5.5 x 2.1 mm centre-positive barrel plug.	R
1	LM2596S buck converter module, FIXED 5 V	Step-down module, input 7-35 V, output FIXED 5.0 V, 3 A or better. Buy the fixed-output version. If only the adjustable pot version exists, it must be set to exactly 5.00 V with a multimeter BEFORE anything is connected.	R

B. SMALL PARTS AND BUILD MATERIALS - buy all of these

QTY	ITEM	EXACT SPECIFICATION	SHOP
10	Resistor 10 kilo-ohm	10k ohm, 1/4 W (0.25 W), +/-5% tolerance, carbon film, axial through-hole (colour code: brown-black-orange-gold).	K
2	Resistor 4.7 kilo-ohm	4.7k ohm, 1/4 W, +/-5%, carbon film, axial through-hole (yellow-violet-red-gold). For the DS18B20 1-Wire pull-up.	K
3	Electrolytic capacitor 1000 uF / 25 V	1000 microfarad, 25 V, aluminium electrolytic, radial leads, 105 C rated preferred. Two used (ESP32 supply + mist board supply) + one spare.	K
5	Glass fuse 3 A fast-blow, 5 x 20 mm	3 A, fast-acting (F3A), glass cartridge, 5 x 20 mm.	K
1	Inline fuse holder, 5 x 20 mm	Waterproof inline holder with wire leads, fits 5 x 20 mm cartridge fuse.	K
2	Stripboard / veroboard, approx. 100 x 160 mm	Copper stripboard, 2.54 mm hole pitch, phenolic or FR4.	K
4	Female pin header strip, 1 x 40	2.54 mm pitch, single row, 40 positions, break-to-length. Two rows of 19 make the ESP32 socket; the rest for sensor plugs.	K
10	PCB screw terminal block, 2-pin	5.0 mm or 5.08 mm pitch, 2 positions, rated 10 A / 300 V, through-hole.	K
6m	Hook-up wire 18 AWG	Stranded, PVC insulated: 3 m red + 3 m black. For adapter, buck and load wiring.	K

	QTY	ITEM	EXACT SPECIFICATION	SHOP
	1	Jumper wire set 22 AWG	Breadboard jumper pack, 65 pcs assorted (male-male, male-female, female-female) or 5 m of 22 AWG stranded in assorted colours.	K / T
	2	DC barrel jack pigtail, female, 5.5 x 2.1 mm	Female jack with approx. 20 cm wire leads, matches the 12 V adapter plug.	K
	1	Heat-shrink tubing assortment	Mixed box, diameters approx. 1.5 mm to 10 mm, 2:1 shrink ratio.	K
	1	Illuminated rocker switch, 12 V	SPST rocker, rated min 6 A at 12 V DC, with built-in 12 V indicator lamp. Master switch for the 12 V line.	K

C. OPTIONAL / FUTURE UPGRADES - buy only if wanted

	QTY	ITEM	EXACT SPECIFICATION	SHOP
	1	DS18B20 waterproof temperature probe	Sealed stainless probe, 1 m cable, 3-wire (VCC GND DATA), -55 to +125 C. Substrate temperature - firmware pin GPIO4 already reserved.	R
	1	SSD1306 OLED display 0.96 inch	128 x 64 pixels, I2C interface, address 0x3C, 4-pin (GND VCC SCL SDA), 3.3 V.	R
	2	DC brushless fan, 12 V, 40 x 40 x 10 mm	12 V DC, 2-wire, sleeve bearing. One exhaust + one circulation.	R
	10	1N5819 Schottky diode	40 V, 1 A, DO-41 axial. Flyback protection - REQUIRED the day fans or a pump are added; not needed for mist modules or LED strip.	R
	1	Active buzzer module, 3.3-5 V	Active type (fixed tone, DC drive), 2-pin or 3-pin module.	T / K
	1	Water level / leak detection module	Bare-trace water sensor board, analog output, 3-pin (VCC GND AOUT). Run at 3.3 V only.	R
	1	Peristaltic dosing pump, 12 V	12 V DC gear-motor peristaltic pump with silicone tubing, approx. 100 mL/min. Precision-watering upgrade.	R
	1	Ultrasonic mist maker, 24 V, with adapter	Pond-type 24 V 0.8 A atomizer with included 24 V 1 A adapter, approx. 350 mL/h. High-output humidity upgrade.	B
	1	ESP32 DevKit V1 - spare board	Identical to the first. Cheap insurance against a wiring mistake.	T / R
	2	Spare atomizer discs, 20 mm	20 mm / 113 kHz ceramic discs matching the 5 V mist modules. The disc is a wear part.	B / D
	1	ESP32-CAM + ESP32-CAM-MB programmer	ESP32-CAM board with OV2640 2 MP camera, plus the CAM-MB micro-USB flashing base.	B / R

Checks at the counter: ESP32 board must say WROOM-32 on the metal shield. BME280 board must be the 3.3 V version. Soil sensors must be capacitive (solid paddle), not resistive (two metal prongs). Test a mist module before leaving: it must mist immediately when powered, with no button press.